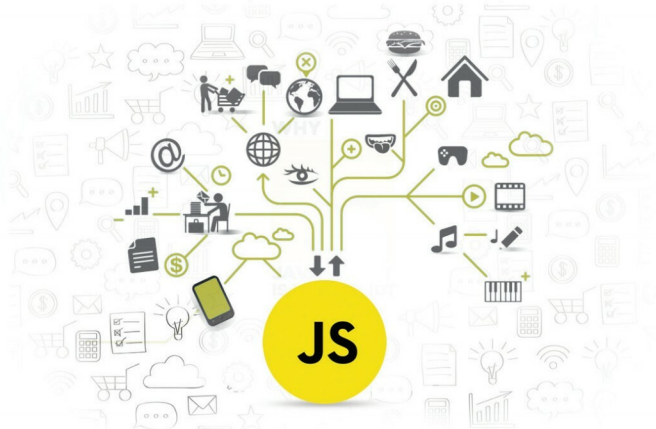


A Peek at Four JavaScript Based Libraries for the Internet of Things

Connecting every 'thing' is the latest trend in the world of IT. A prime challenge in connecting heterogeneous things is the need for interoperable development tools. This article introduces readers to a few important JavaScript based libraries, frameworks and platforms associated with IoT development.



The Internet has definitely changed the way we carry out our day to day activities, be it something as simple as communicating with a friend, or the more critical applications such as banking and healthcare. The benefits of connecting more and more devices to the Internet, popularly known as the Internet of Things (IoT), are manifold. As an example, your keychain can broadcast its location, helping you to find the key you lost by clicking on an app on your smartphone.

However, as more and more things get connected, there are new requirements in network protocols, security, programmability, etc.

JavaScript and IoT

JavaScript has become the *de-facto* client-side scripting language of the Web. One can call JavaScript the 'C of

the Web'. There are millions of Web developers with JavaScript skills who can migrate with relative ease to IoT. Moreover, with the exponential growth in IoT, there are specialised JavaScript frameworks that can use the potential of connected devices.

JavaScript can be utilised for IoT in the following different ways (Figure 1):

- One of the approaches is to run JavaScript in the host computer and send the signals to the client (things). This mode is optimal for scenarios in which the actual 'things' don't have the potential to run even the slimmest JavaScript codebase.
- Another approach is to execute the JavaScript code through memory optimised engines in the device itself. Frameworks such as JerryScript may be used to run the devices.